Product Price Monitor

Machine Learning Analysis Report

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Executive Summary

This report summarizes automated price monitoring across 169 tracked products over 14 scrape runs. Two machine learning techniques were applied to the collected data: anomaly detection, to identify products priced unusually relative to their category, and price forecasting, to project short-term price movement based on historical trend. The analysis identified 24 products with pricing that falls outside the expected range for its category. Among products with sufficient history, 1 show an upward price trend and 0 show a downward trend. Based on 14 scrape runs — sufficient history for a reasonably confident short-term linear projection.

Anomaly Detection

Each product's latest price is compared against the mean and standard deviation of its own category, using a Z-score threshold of 1.5 standard deviations. To prevent a single extreme value from distorting its own baseline, each product is excluded from the calculation of its category's mean and standard deviation before being scored against it. This avoids a common statistical pitfall where an outlier masks itself by pulling the average toward its own value.

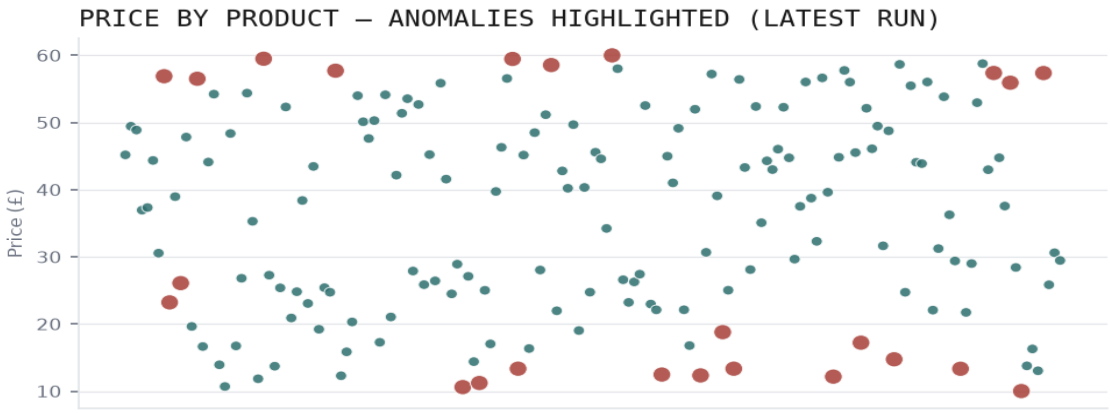


Figure 1. Price by product for the latest scrape run. Each point represents one product; flagged anomalies are shown larger and in a distinct color.

Across the 5 tracked categories, price spread varies considerably: Fiction shows the widest range in this run, from £10.60 to £59.98. The most significant anomaly is “A Year in Provence (Provence #1)”, priced at £56.88 against a Travel category average of £39.79 — a Z-score of 2.05, placing it above the expected range for its category.

Product	Category	Price	Category Avg	Z-Score	Direction

A Year in Provence (Proven…	Travel	£56.88	£39.79	2.05	Above
Every Heart a Doorway (Eve…	Fantasy	£12.16	£39.59	-2.04	Below
The Road to Little Dribbli…	Travel	£23.21	£39.79	-1.96	Below
Sister Sable (The Mad Quee…	Fantasy	£13.33	£39.59	-1.95	Below
Princess Between Worlds (W…	Fantasy	£13.34	£39.59	-1.95	Below
Harry Potter and the Chamb…	Fantasy	£14.74	£39.59	-1.84	Below
Boar Island (Anna Pigeon #…	Mystery	£59.48	£31.72	1.81	Above
I Am Pilgrim (Pilgrim #1)	Fiction	£10.60	£36.07	-1.75	Below
Dear Mr. Knightley	Fiction	£11.21	£36.07	-1.71	Below
The No. 1 Ladies' Detectiv…	Mystery	£57.70	£31.72	1.69	Above
Immunity: How Elie Metchni…	Science	£57.36	£33.09	1.65	Above
The Disappearing Spoon: An…	Science	£57.35	£33.09	1.65	Above
A Feast for Crows (A Song …	Fantasy	£17.21	£39.59	-1.64	Below
Last One Home (New Beginni…	Fiction	£59.98	£36.07	1.64	Above
The Silent Wife	Fiction	£12.34	£36.07	-1.62	Below
Lila (Gilead #3)	Fiction	£12.47	£36.07	-1.62	Below
The Past Never Ends	Mystery	£56.50	£31.72	1.60	Above
The Improbability of Love	Fiction	£59.45	£36.07	1.60	Above
The Origin of Species	Science	£10.01	£33.09	-1.56	Below
Mothering Sunday	Fiction	£13.34	£36.07	-1.55	Below
Miller's Valley	Fiction	£58.54	£36.07	1.54	Above
The Fabric of the Cosmos: …	Science	£55.91	£33.09	1.54	Above
1,000 Places to See Before…	Travel	£26.08	£39.79	-1.53	Below
Unicorn Tracks	Fantasy	£18.78	£39.59	-1.52	Below

Table 1. All products flagged as anomalous in the latest run, ranked by absolute Z-score. A positive Z-score indicates a price above the category average; a negative value indicates a price below it.

Price Forecast

A linear regression model is fit to each product's price history, using scrape run sequence as the independent variable, to project price for the next scheduled run. This is a simple trend-following method: it assumes the rate of change observed so far will continue, and does not account for external market factors. Based on 14 scrape runs — sufficient history for a reasonably confident short-term linear projection.

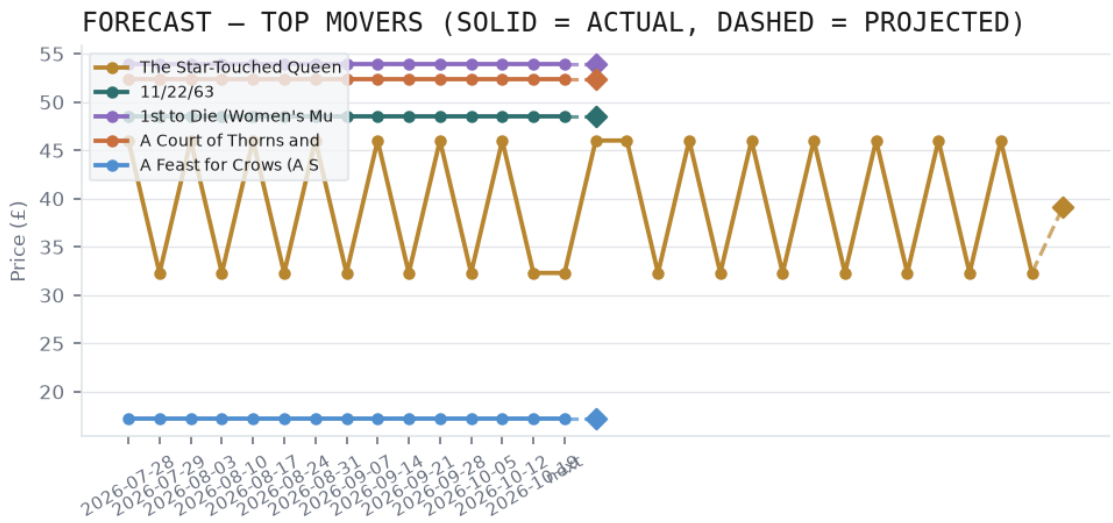


Figure 2. Actual price history (solid lines) and projected next-period price (dashed lines, diamond marker) for the five products with the largest projected percentage change.

The product with the largest projected movement is “The Star-Touched Queen”, currently priced at £32.30 and projected to increase to approximately £39.16 (+21.2%) by the next scrape run. This projection is based on 14 historical data points per product and reflects the linear trend observed to date.

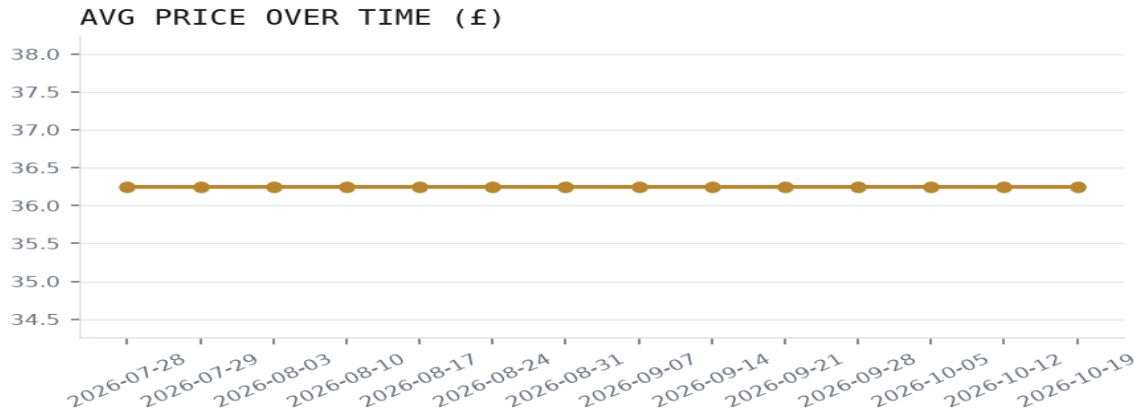
Product	Current Price	Forecast Price	Change	Trend
The Star-Touched Queen	£32.30	£39.16	21.2%	Up
11/22/63	£48.48	£48.48	0.0%	Flat
1st to Die (Women's Murder C…	£53.98	£53.98	0.0%	Flat
A Court of Thorns and Roses …	£52.37	£52.37	0.0%	Flat
A Feast for Crows (A Song of…	£17.21	£17.21	0.0%	Flat

Table 2. The five products with the largest projected price change, ranked by absolute percentage change.

Reference Charts

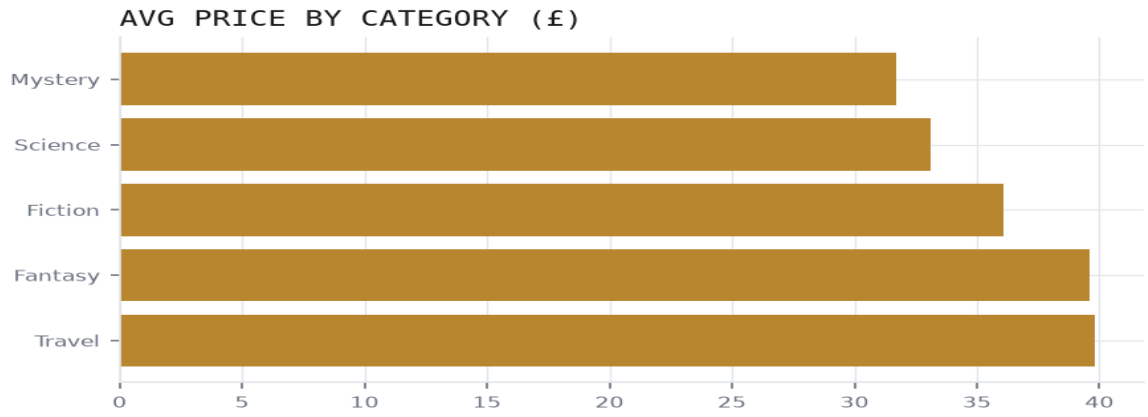
The charts in this section summarize the underlying dataset used in the analysis above and are provided for additional context.

Average Price Over Time



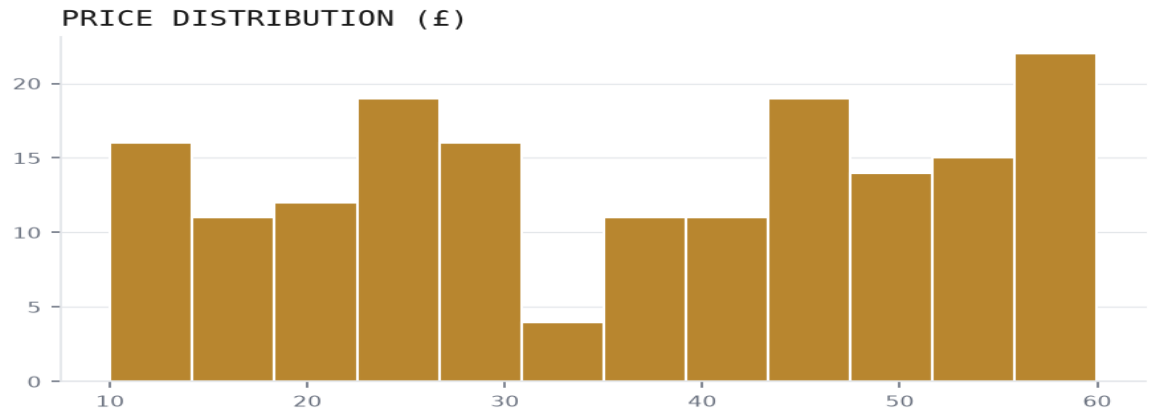
Average price across all tracked products currently stands at £36.24. Since the first recorded run, the overall average has stayed flat by £0.00 (+0.0%).

Average Price by Category



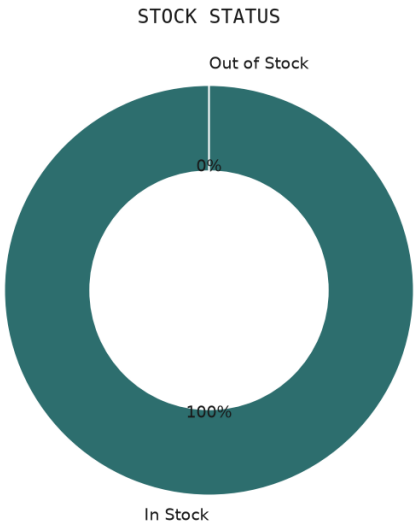
Travel carries the highest average price in this run, at £39.79, while Mystery is the lowest, at £31.72.

Price Distribution



Prices in the latest run range from £10.01 to £59.98, with a median of £37.97. The distribution below shows how tracked products are spread across that range.

Stock Availability



Of the 169 products tracked in the latest run, 170 are currently in stock and 0 are out of stock.